

Worksheet:

WorkSheet 1

1. $y = e^{2x}$

2. $f(x) = \sin e^{3x}$

3. $y = e^{\sin(x^4)}$

4. $f(t) = \frac{1}{t^4}$

5. $y = (t - t^3)^{10}$

6. $f(t) = \cos^4 3t$

7. $f(x) = 2^x$

8. $f(x) = 2^{\operatorname{cosec}(8x)}$

9. $f(x) = \cot(2^{1-5x})$

10. $f(t) = 7 \tan(3t) - 3^{\tan 7t}$

11. $f(x) = \operatorname{cosec} \frac{\pi}{3}$

12. $f(x) = \frac{\sqrt{x^3+1}}{7 - \tan 7x}$

Worksheet on differentiation - sum, product, quotient and chain rules 1

Find the derivatives of the following functions. DO NOT simplify your answer

1. $y = \left(\frac{1+x}{x^3} \right)^4$

2. $y = \frac{1}{(2-3x)^3}$

3. $f(t) = \cot(2^t)$

4. $f(t) = \tan(e^t) \cdot e^{\tan t}$

5. $f(v) = \sqrt{v^3 + \sec v}$

6. $f(s) = \frac{1}{1-3s}$

7. $f(x) = 2 \cos(e^{-4x})$

8. $f(x) = e^{\sin 4x} \cdot \cos(2x)$

9. $f(x) = 2^{\tan 3x}$

10. $f(x) = 5 \sec(3^x)$

11. $f(x) = \operatorname{cosec} \frac{\pi}{3}$

12. $f(x) = (1 + 2 \sin 5x)^3$

13. $f(x) = 3^{5x} + 5x^3 + 5^{3x}$

14. $f(x) = 5(1-2x)^{10}$

15. $f(x) = \frac{4 \sin x^4}{e^{4 \sin x}}$

16. $y = \sqrt{\cos(e^{3x+1})}$

17. $y = 2e^{\cot x} - 3 \cot(e^{-x})$

18. $f(x) = \sqrt[3]{\sec(4x)} \tan(x^4)$

19. $f(x) = \frac{\sqrt{x^3+1}}{7-\tan 3x}$

20. $y = \frac{2^{3x} \sin(4x)}{\cos(1-x) \operatorname{cosec}(e^{-x})}$